

```
In [ ]: #I want to find a qube of numbers
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```
In [3]: lst = [1,2,3,4,5,6,7]
```

```
In [4]: for i in lst:  
        print(i*i*i)
```

```
1  
8  
27  
64  
125  
216  
343
```

```
In [5]: def cube(i):  
        return(i*i*i)
```

```
In [8]: cube(10)
```

```
Out[8]: 1000
```

```
In [9]: cub_result = []  
        for i in lst:  
            cub_result.append(cube(i)) #cube(i) is a functions, it is performaing cube operation  
        print(cub_result)
```

```
[1, 8, 27, 64, 125, 216, 343]
```

```
In [ ]: #Map  
#map is function in python  
#map syntax -- map(function, list/tuple)  
#map is a builin function that is used to apply a function to each element of list/tuple
```

```
In [13]: cub_num = map(cube,lst)  
         print(list(cub_num))
```

```
[1, 8, 27, 64, 125, 216, 343]
```

```
In [19]: numbers = [1,2,3,4,5,6,7,8,9,10,11,12]
```

```
In [15]: def even_num (n):  
        if n % 2 == 0:  
            return n
```

```
In [18]: even_num(2)
```

```
Out[18]: 2
```

```
In [25]: even_num_output = []
         for i in numbers:
             if even_num(i):
                 even_num_output.append(i)
         print(list(even_num_output))
```

[2, 4, 6, 8, 10, 12]

```
In [ ]: #filter
         #filter syntax (function, list/tuple)
         #filter is a built in function that can be used to filter the iterable elements (we ne
         #function will be applied to every element also gets filtered according to logic.
```

```
In [28]: even_nu = filter(even_num,numbers)
         print(list(even_nu))
```

[2, 4, 6, 8, 10, 12]

```
In [ ]: #Voter age check
```

```
In [29]: voter_age_list = [12,22,17,18,10,25,27]
```

```
In [30]: def voteragecheck(n):
         if n >=18:
             return(n)
```

```
In [32]: voteragecheck(22)
```

Out[32]: 22

```
In [34]: voter_age_check = [] #defining empty list
         for i in voter_age_list:
             if voteragecheck(i):
                 voter_age_check.append(i)
         print(list(voter_age_check))
```

[22, 18, 25, 27]

```
In [36]: voterage_check = filter(voteragecheck,voter_age_list)
         print(list(voterage_check))
```

[22, 18, 25, 27]

```
In [ ]: #oddnumber check
```

```
In [37]: listofnumbers = [1,2,5,7,99,4,3,4,66,77]
```

```
In [40]: def oddnumcheck (n):  
         if n % 2 != 0:  
             return(n)
```

```
In [42]: oddnumcheck(5)
```

```
Out[42]: 5
```

```
In [44]: oddnumbercheck = filter(oddnumcheck,listofnumbers)  
         print(list(oddnumbercheck))
```

```
[1, 5, 7, 99, 3, 77]
```

```
In [ ]: #map use case #exmaple - Let's say I need to implement salary hike 20%
```

```
In [45]: salary = [12000,10000,7000,15000,17000,20000]
```

```
In [48]: def salarihike(n):  
         return(n + (n * 0.2))
```

```
In [51]: salaryhike1 = map(salarihike,salary)  
         print(list(salaryhike1))
```

```
[14400.0, 12000.0, 8400.0, 18000.0, 20400.0, 24000.0]
```

```
In [ ]: #filter example filtering the Names starting with 'S'
```

```
In [52]: names = ["Santosh","Suparna Babu","Rupesh","Praneeth","SriRam","Sayed","Sabu","Ranjita"]
```

```
In [54]: def snames(nm):  
         if nm[0] == "S":  
             return nm
```

```
In [56]: snames("Santosh")
```

```
Out[56]: 'Santosh'
```

```
In [63]: NamesstartingwithS = filter(snames,names)  
         print(list(NamesstartingwithS))
```

```
['Santosh', 'Suparna Babu', 'SriRam', 'Sayed', 'Sabu']
```